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MILESTONE ONE

Prepared for:

Software Engineering Team

IITM BS Online Degree

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FANTASTIC FOUR

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IIT MADRAS

Identification of Users

PRIMARY USERS

Learners / Students

Who they are:

Students, interns, and faculty participants enrolled in VLED programs who seek information regarding courses, internships, attendance policies, cohort details, and program logistics.

Why they are primary users:

- They are the main stakeholders who directly interact with the system to seek program-related information.
- Their queries initiate the entire support workflow, making them central to the system's purpose.
- The effectiveness of the solution is measured by how efficiently it resolves their doubts.
- Their engagement , feedback, and usage determine the success and continuous improvement of the platform.

SECONDARY USERS

Support Desk / Customer Support Staff

Who they are:

Operational staff responsible for resolving learner queries, responding to tickets, and maintaining support SLAs.

Why they are secondary users:

- They intervene when automated responses are insufficient, ensuring unresolved queries are addressed.
- They are responsible for maintaining service timelines and ensuring learner concerns are handled promptly.
- Their responses contribute to expanding and refining the system's knowledge base.
- They depend on the platform to streamline repetitive queries and manage operational workload efficiently.

TERTIARY USERS

Admins / Program Managers / Lab Coordinators

Who they are:

Faculty coordinators and lab heads responsible for policy decisions and oversight.

Why they are tertiary users:

- They oversee the accuracy, relevance, and professionalism of the information provided through the system.
- They monitor patterns in user queries to identify areas requiring policy clarification or updates.
- They ensure that institutional guidelines and program changes are consistently reflected in responses.
- They use system-level insights to make informed decisions about program communication and operational improvements.

User Research

RESEARCH METHOD

To identify user requirements, we conducted:

- A structured interaction with a VLED representative to understand operational challenges in managing learner queries.
- Observation of the current workflow involving website FAQs, Discord discussions, and support ticket handling.
- Limited web-based research on how online education platforms manage large-scale learner support.

The focus was on identifying operational pain points rather than suggesting features.

A) Key Findings from Interview and Observation

1. **High Volume of Repetitive Queries**
   Many support tickets contain questions already addressed on the website or FAQs, but learners prefer direct clarification instead of navigating static resources.

2. **Fragmented Communication Channels**

   Information is spread across website content,FAQs,Discord discussions, and manual ticket responses, leading to inconsistency.

3. **Time-Sensitive Support Requirements**

   Support staff are expected to respond within a short time frame (e.g., 2 hours),increasing operational pressure during peak periods.

4. **Lack of Dynamic Updates**

   Policy changes or cohort updates are not automatically reflected across all communication channels.

5. **Limited Feedback and Analytics**

   There is no structured mechanism to track learner feedback or analyse frequently asked topics.

B) Web-Based Observations

A brief study of digital education platforms indicates that many use:

- Conversational AI interfaces to reduce dependency on static FAQs.
- Escalation systems when automated responses fail.
- Analytics dashboards to track query trends.

However, challenges such as maintaining updated information, ensuring consistency, and managing peak query loads remain common.

Research Summary

The findings confirm the need for a centralised, dynamically updated, and monitored query management system with human oversight. These insights form the basis for the user stories defined in this milestone.

User Stories

Primary User Stories – Learners

1. **As a learner** enrolled in a VLED program,
   I want to ask questions through an interactive chat interface,
   So that I can receive immediate clarification without searching through multiple webpages.

2. **As a learner**,
   I want the system to provide responses that reflect the most recent program policies and cohort details,
   So that I do not rely on outdated or conflicting information.

3. **As a learner,**
   I want the system to inform me when my query cannot be answered automatically,
   So that I can escalate it to the support team without uncertainty.

4. **As a learner,**
   I want to provide feedback on responses I receive,
   So that unclear or inaccurate information can be reviewed and corrected.

5. **As a learner,**
   I want consistent answers regardless of whether the source is website content, FAQ documentation, or Discord discussions,
   So that I do not experience confusion due to variation in communication channels.

6. **As a learner,**
   I want frequently asked questions to be resolved instantly,
   So that I can focus on learning activities instead of waiting for support replies.

Secondary User Stories – Support Staff Desk

7. **As a support desk staff member,**
   I want only unresolved or newly generated queries to be routed to me,
   So that repetitive questions are handled automatically and my workload remains manageable.

8. **As a support desk staff member,**
   I want to respond to escalated queries through an internal dashboard,
   So that my responses are stored and reused for similar future queries.

9. **As a support desk staff member,**
   I want to view pending tickets along with submission timestamps,
   So that I can ensure responses are delivered within the defined response window (e.g., 2 hours).

10. **As a support desk staff member,**
    I want visibility into daily query volume and resolution status,
    So that I can plan resource allocation during peak periods.

Tertiary User Stories – Admins / Program Managers

11. **As an admin**,
    I want to review and approve answers added to the knowledge base,
    So that institutional standards and tone are maintained.

12. **As an admin,**
    I want to track frequently asked topics and areas generating negative feedback,
    So that I can identify policy gaps or communication ambiguities.

13. **As an admin,**
    I want system responses to automatically reflect updates made on the official website,
    So that learners always receive synchronized and accurate information.

14. **As an admin,**
    I want access to performance analytics such as query trends and feedback ratios,
    So that I can evaluate the effectiveness of the support system.

AI Usage Section

Description of AI Usage

AI tools were used as a research and documentation support aid during this milestone.

Specifically, the GPT Deep Research tool was used to:

- Study how online education platforms handle large volumes of learner queries.
- Understand how similar systems implement AI-based support, escalation workflows, and knowledge management.
- Learn best practices for writing user stories aligned with SMART guidelines.

We also used Grammarly to auto-correct the wrong sentences and provide grammar correction and frame better academic style sentences.

All project requirements were derived from stakeholder interaction; AI was used only to assist in structuring and refining the documentation.

AI Interaction Reference:

CHATGPT

Advantages of Using AI

- Faster access to structured information on comparable online education services.
- Assistance in organizing interview findings into clear user stories.
- Context-aware guidance for improving clarity and alignment with academic guidelines.

Challenges Faced

- Refining prompts to obtain context-specific rather than generic responses.
- Ensuring AI-generated suggestions accurately reflected the stakeholder's actual requirements.
- Critically reviewing outputs to maintain academic integrity and relevance.